

Quiz 2: Disaster Management (CE8.401) - Solutions

Q 1: Explain the following cognitive biases and how they affect community response during an extreme event like Typhoon Haiyan: a) Normalcy Bias, b) Optimism Bias (2 marks)

- Normalcy Bias : This is the tendency for people to believe that because a disaster has never happened to them before, it won't happen now . During Typhoon Haiyan, this caused many to stay in their homes because they expected "just another storm" rather than a catastrophic event . (1 mark)
- Optimism Bias : This is the "it won't happen to me" mindset. People acknowledge the risk exists for others but believe they are personally safe or that their house is stronger than their neighbor's, leading to a delay in evacuation.(1 mark)

Q 2: Use the Swing Analogy to explain the behaviour of unsymmetric buildings (mass vs. stiffness unsymmetric) during ground shaking (2 marks)

- When one rope is much shorter than the other, or the seat is weighted heavily on one side.
- When you push it, it won't just go straight back and forth; it will start to twist (torsion).
- In an unsymmetric building, the Center of Mass and Center of Stiffness don't line up. During an earthquake, the building twists instead of just swaying, which puts massive stress on the corners and can lead to a quick collapse .

Q 3: Contrast the structural behavior of buildings under Wind Force versus Earthquake Effects in terms of elasticity and inelasticity (2 marks)

- We design buildings to stay elastic under wind loads. This means the building can sway, but it should return to its original shape without any permanent damage.
- Earthquakes carry so much energy that it's too expensive to make buildings perfectly elastic. Instead, we design them to be inelastic (ductile). This means we allow the building to undergo controlled damage (like cracks in specific beams) to soak up the earthquake's energy, which prevents the whole building from falling down .

Q 4: Explain Butterfly Effect in the context of weather forecasting models. Why is there a theoretical limit of roughly 2 weeks for accurate forecasts? (2 marks)

- This is the idea that the atmosphere is so sensitive that a tiny change (like a butterfly flapping its wings) can cause a massive change in the weather somewhere else weeks later.
- Because our initial data (temperature, pressure, etc.) is never 100% perfect, those tiny errors grow bigger every day due to the Butterfly Effect . By about 14 days, the errors become so huge that the forecast is no better than a random guess.

Q5: Differentiate between the utility of short-range forecasts and long-range climate projections in disaster management (2 marks)

- Short-range Forecasts: These are for immediate action. They tell us if we need to evacuate a city tomorrow or open a dam's floodgates right now.

- Long-range Projections: These are for planning and policy. They help us decide where to build new cities, how high to build sea walls for the next 50 years, and how to manage water resources for future droughts.

Q6: Compare Drip Irrigation and Sprinkler Irrigation as on-farm water management techniques for drought mitigation (2 marks)

- Drip Irrigation: Water is delivered drop-by-drop directly to the roots. It is the "gold standard" for drought because it has almost zero waste from evaporation or wind.
- Sprinkler Irrigation: This sprays water into the air like rain. It's good for covering large areas, but in a drought, it's less efficient because a lot of water evaporates before it even hits the ground.

Q7 (a): Define Duty (D), Delta (Δ) and Base period (B). State the mathematical relationship between them (2 marks)

- Duty (D): The area (in hectares) that 1 cumec of water can irrigate if it flows constantly during the entire growing season.
- Delta : The total depth of water (in meters) the crop needs from start to finish.
- Base Period (B): The total time (in days) between the first watering for sowing and the last watering before harvest.
- Relationship: $\Delta = 8.64 \text{ B/D}$

Q7 (b): A specific variety of wheat has a base period of 120 days.

If the total depth of water required for its full growth is 40 cm, calculate the duty of the water for this crop (2 marks)

B = 120 days , Delta = 40cm

D = 2592 hectares/cumec

Question 8 (a): Do you think cloud seeding is a feasible strategy for drought mitigation?

Justify your answer based on the scientific constraints of the technology (2 marks)

Cloud seeding is not a standalone solution for drought. The biggest scientific constraint is that it cannot create" clouds; it only helps existing clouds rain more efficiently. If there are no moisture-rich clouds because of a severe drought, cloud seeding will not work.. Also this is not well scientifically validated

Question 8 (b): Compare the scientific and operational feasibility of cloud seeding for Region Alpha and Region Beta. Which region is more likely to see a successful increase in precipitation, and why? (2 marks)

Region Alpha has the right ingredients (high humidity and dense clouds), but the terrain is a nightmare for setup. Region Beta has great infrastructure and easy access, but it lacks the ingredients (it has thin, dry clouds).

Best : Region Alpha. Even though it's harder to get there, cloud seeding is a scientific process first. Without the high humidity and strong updrafts found in Alpha, all the equipment in the world in Region Beta won't produce significant rain.

Q9. Propose a comprehensive digitally integrated Disaster Management system for a flood-prone district in India. Your design should cover the full Disaster Management cycle: Preparedness, Response, Recovery, and Mitigation. Clearly specify the digital technologies used at each stage. Also identify at least two limitations or failure points in your proposed system and suggest remedial measures for them. (5 marks)

Digitally Integrated Disaster Management System:

Preparedness

(1 marks for any 2 points)

- Remote Sensing & GIS: Satellite imagery maps flood-prone zones, identifies vulnerable populations, and creates hazard layers (as used in NDEM).
- AI/ML Predictive Models: Trained on historical flood, rainfall, and river level data to forecast flood probability, intensity, and affected zones, particularly for short term warnings.
- IoT Environmental Sensors: Deployed along riverbanks and reservoirs to continuously monitor water levels, rainfall, soil moisture, and drainage capacity.
- Digital Twin: A virtual model of the district's river network and urban infrastructure simulates flood scenarios to test preparedness strategies.
- Community Apps: Mobile apps (like M-Kavach) educate residents on evacuation routes, emergency contacts, and safety protocols.

Response

(1 marks for any 2 points)

- Automated Alert System: When sensors reach their thresholds, SMS/voice alerts and push notifications are automatically sent to residents- similar to M-Kavach.
- Drones: Provide real-time aerial imagery of flooded areas, locate stranded survivors, and guide rescue teams efficiently.
- Crowdsourcing & Social Media Monitoring: NLP algorithms scan Twitter/Facebook for distress signals, helping identify areas needing immediate attention.
- Centralized Dashboard: Emergency responders access a real-time GIS dashboard (like IDRN) showing available resources — boats, personnel, medical supplies — at district and state levels.
- Blockchain for Aid: Ensures transparent, tamper-proof distribution of relief materials to affected families.

Recovery

(1 marks for any 2 points)

- AI Damage Classification: Satellite and drone imagery processed through image recognition to classify damage as minor, moderate, severe, or collapse — enabling prioritization.
- BIM (Building Information Modelling): Tracks reconstruction progress digitally.
- Digital Databases: Manage compensation claims, insurance records, and aid distribution transparently.
- Tele-medicine Platforms: Provide remote healthcare (like Markeshift hospitals) and temporary shelters to displaced populations when physical infrastructure is damaged.

Mitigation*(1 marks for any 2 points)*

- Infrastructure Vulnerability Assessment/Analysis: ML models assess structural weaknesses in bridges, embankments, and buildings in flood-prone areas.
- Land Use Planning: GIS models guide planners to keep critical infrastructure away from high-risk flood zones.
- Smart Infrastructure: IoT sensors in dams and embankments monitor structural health continuously, alerting authorities before failure.
- Climate Change Modelling: Hydrological models predict long-term flood risk patterns under different climate scenarios.

Limitations:*(1 marks)**Other limitations may also be considered*

Limitation	Remedial Measure
Connectivity failure: local towers and internet go down during severe floods, disabling alerts and dashboards.	Deploy satellite communication as backup; pre-position offline-capable devices with local responders.
Sensor data gaps: IoT sensors may malfunction or give false readings under extreme conditions	Use redundant sensor networks with cross-validation via satellite imagery and AI anomaly detection to filter unreliable data